

colour separation. If you start with an RGB image, it's best to edit first in RGB and then convert to CMYK at the end of your process. In RGB mode, you can use the **Proof Setup** commands to simulate the effects of a CMYK conversion without changing the actual image data. (See *Soft-proofing colours*.) You can also use CMYK mode to work directly with CMYK images scanned from high-end systems.

Although CMYK is a standard colour model, the exact range of colours represented can vary, depending on the press and printing conditions. Photoshop's CMYK Colour mode varies according to the working space setting that you have specified in the **Colour Settings** dialog box.

1.4.3. Lab Colour Mode

In Photoshop, the Lab Colour mode has a lightness component (L) that can range from 0 to 100. In the Adobe Colour Picker, the 'a' component (green-red axis) and the 'b' component (blue-yellow axis) can range from +127 to -128. In the Colour palette, the 'a' component and the 'b' component can range from +120 to -120.

You can use the Lab mode to work with Photo CD images, edit the luminance and the colour values in an image independently, move images between systems, and print to PostScript Level 2 and Level 3 printers. To print Lab images to other colour PostScript devices, convert to CMYK first.

Lab images can be saved in Photoshop, Photoshop EPS, Large Document Format (PSB), PDF, Photoshop Raw, TIFF, Photoshop DCS 1.0 or Photoshop DCS 2.0 formats. 48-bit (16 bits per channel) Lab images can be saved in Photoshop, Large Document Format (PSB), Photoshop PDF, Photoshop Raw or TIFF formats.

Note: The DCS 1.0 and DCS 2.0 formats convert the file to CMYK when opened.

Lab colour is the intermediate colour model Photoshop uses when converting from one colour mode to another.